

Chapter 11: The "Cloud" (Collaboration & Hosting)

I. Version Control and The Cloud

- **The Problem:** The absolute wrong way to collaborate on code is by _____ files back and forth.
- **The Solution:** A _____ (VCS), with _____ being the most popular system.
- **Git's Role:** Git acts as a "_____" allowing you to rewind the project back to the last working version if a bug is introduced.

Cloud Computing

- **Definition:** Cloud Computing is simply _____ from someone else over the internet.
- **Physical Location:** "The Cloud" is just a nickname for giant warehouses called _____ filled with powerful computers (**Servers**).
- **Web Hosting:** The service of storing your finalized web application files on a _____ so they can be accessed globally 24/7.

The Big Three Cloud Giants

Platform	Analogy	Primary Strength/Integration
AWS	The "_____" of the cloud.	The Original & The Biggest (hosts Netflix).
Azure	The "_____" of the cloud.	Integrates heavily with Windows/Office and large enterprise businesses.
GCP	The "_____" King.	Famous for powerful data analysis and AI tools (hosts Google Search/YouTube).

Deep Dive Discussion

1. **Collaboration Chaos:** Why is using Git essential when two developers are working on the

same HTML file at the same time? What specific problem does it solve?

2. **Hosting Necessity:** Why can't a professional website be hosted on a standard school computer? Name two reasons related to speed or reliability.

Application Activity

1. **Team Problem:** A teammate just emailed you the latest version of the styles.css file, but you suspect their changes broke the site. How would using a **VCS** (like Git) allow you to test their changes safely without overwriting your own file?

II. Cloud Service Models (SaaS, PaaS, IaaS)

- **Definition:** These models define _____ you want to rent and control.

Model	Full Name	Analogy	Your Responsibility
SaaS	_____ _____	You just use the finished software.	_____ (just using the app).
PaaS	_____ _____	The provider gives you the kitchen (environment and tools).	Writing and managing your _____ code.
IaaS	_____ _____	The provider gives you the empty warehouse (raw hardware).	Installing the OS, managing the network, and installing your application.

Deep Dive Discussion

1. **PaaS vs. IaaS:** You are deploying a simple, static website (HTML/CSS/JS). Why would **PaaS** (like Netlify or Vercel) be a much better, cheaper, and faster choice than **IaaS** (renting a raw virtual server)?
2. **SaaS Examples:** Name two real-world cloud applications you use every day that fit the **SaaS** model.

Application Activity

1. **Model Identification:** Match the following scenarios to the correct Cloud Service Model

(SaaS, PaaS, or IaaS):

- **A.** Renting a virtual server from AWS, then spending two hours installing the operating system and web server software. _____
- **B.** Logging into Google Docs to collaborate on a school paper. _____
- **C.** Using a simple hosting service that automatically manages the server environment after you upload your code. _____

III. Deployment Models and Scalability

Cloud Deployment Models

Model	Analogy	Description	Key Feature
Public Cloud	Renting an _____ in a giant building.	Run on shared hardware owned by a major provider.	Best _____ and lowest cost.
Private Cloud	Owning a private _____.	Dedicated infrastructure for a single organization.	Maximum _____ and control.
Hybrid Cloud	A mix of the two models.	Sensitive data stored privately; public website hosted publicly.	Combines the security of _____ with the scalability of _____.

The Cloud's Superpower

- **Scalability:** The ability of a system to easily handle a growing amount of work or _____.
- **Elasticity:** The dynamic growth and shrinking of resources, allowing the system to _____ resources instantly during a traffic spike and _____ them when the spike ends.

Deep Dive Discussion

1. **Traffic Spike:** A viral video just linked to your small personal website, causing 100,000 visitors in an hour. How does the cloud's **Elasticity** save your site from crashing and save you money afterward?

2. **Hybrid Use Case:** Why would a major bank choose a **Hybrid Cloud** model over a purely Public Cloud model?

Application Activity

1. **Service Model Summary:** Define the three cloud service models (SaaS, PaaS, and IaaS) in your own words, focusing on who is responsible for managing the **Server/OS** and who is responsible for managing the **Application Code**.